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The Computable Protocol: *Revisited*

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Abstract

Clinical trial protocols have long served as the authoritative description of study design and execution, yet they remain largely document-based artifacts. They are written to be read and interpreted by experts, but not necessarily to support systematic evaluation, reuse, comparison, or downstream execution. As a result, many critical design assumptions remain embedded in narrative text until late in the development process, when changes are more disruptive and costly.

The computable protocol is not a new idea. For more than two decades, industry groups, researchers, standards organizations, and regulators have explored ways to represent protocol content in structured, machine-processable forms. Early work anticipated many of the capabilities still being pursued today. Those efforts did not become routine practice, not because the vision was wrong, but because clinical development lacked the global alignment, technical infrastructure, and implementation maturity needed to support broad adoption. This paper revisits that earlier computable protocol vision in light of ICH M11 and the renewed path it creates for structured protocol content.

ICH M11 changes the practical conditions for this discussion. M11 does not invent computable protocols, trial simulation, synthetic populations, or advanced analytics. Its importance is that it provides a globally harmonized framework for organizing and describing core protocol content in a way that can support both human readability and downstream machine processing. Its greater value lies in enabling structured protocol content to become a reusable study design asset rather than a more consistently formatted document.

When protocol content is computable and connected to an underlying data model, study teams can evaluate design assumptions, test alternatives, and understand key dependencies before the trial is set in motion. Different disciplines can work from the same underlying study design while viewing the relationships most relevant to their responsibilities.

The promise is simple: know more before the trial begins.

1. Protocol at the Center of Trial Design and Execution

The clinical trial protocol is the primary reference point for study design and execution. It defines the study population, treatment strategy, assessments, endpoints, statistical considerations, safety monitoring, and operational procedures that shape how a trial is conducted, analyzed, and reviewed. Clinical operations teams use protocol content to develop execution plans. Data managers translate protocol requirements into data collection instruments and edit checks. Statisticians prepare analysis plans. Clinical pharmacologists assess dosing and sampling strategies. Medical monitors and safety physicians define oversight expectations. Investigators and clinical research coordinators use the protocol to guide site conduct, including subject recruitment, study procedures and data collection. IRBs and ethics committees use the protocol to evaluate participant protection, risk-benefit, informed consent, and whether the planned study procedures are appropriate for the proposed population. CROs operationalize the protocol across sites, systems, monitoring activities, and data collection processes. Regulators use the protocol to evaluate whether the study design can support the sponsor's intended conclusions.

Despite this role, protocols have historically been created, reviewed, and implemented primarily as narrative documents. That document-centric approach has important limitations. Narrative protocols can be read and interpreted, but they are difficult to evaluate systematically. They may be internally repetitive, key information may be distributed across multiple sections, and design assumptions may be implicit rather than explicit. Changes made in one section may also affect other sections in ways that are not immediately visible.

As a result, many protocol questions are still addressed through manual review, meetings, spreadsheets, separate feasibility assessments, and functional interpretation. The problem is not that experts lack judgment. The problem is that experts often must work from protocol information that is not organized in a way that supports systematic evaluation of study design choices.

The practical questions are concrete. If an eligibility criterion is tightened, what happens to the eligible population and recruitment timeline? If an assessment is added at Week 10, what happens to participant burden, site workload, cost, and data collection requirements? If an endpoint moves from Week 24 to Week 36, what protocol elements, analysis assumptions, and downstream artifacts are affected? In a document-centric environment, these questions are answered through manual interpretation and separate analyses. In a computable protocol environment, the structured protocol can become the reference point for asking and tracing them.

A computable protocol makes that content more visible, testable, and connected before the trial begins and before downstream processes depend on it.

2. Evolution of the Computable Protocol

The vision of the computable protocol has a long history. In 2002, Michael Kahn described computable protocol models for interactive trial design. That work recognized that clinical trial protocols contain complex relationships among study events, treatments, patient safety requirements, and data collection tasks. It proposed formal computable models to represent protocol and study concepts, identify ambiguities, assess operational risk, support simulation, and configure downstream execution tools.

That early work remains important because it shows that the core idea is not new. The industry has known for decades that protocol content could be represented in structured forms and used to support better design, execution, and analysis. Subsequent standards initiatives continued this evolution. The CDISC Protocol Representation Model, developed collaboratively by CDISC, NIH/NCI, FDA and HL7, defined standard protocol elements and contributed to structured representation of protocol content. FDA and HL7 explored structured protocol exchange and computable study design representations. The FDA/NIH Clinical Trial Protocol Template and the TransCelerate Common Protocol Template added another important phase: protocol harmonization, with efforts focused on improving protocol completeness, consistency, common section structure, common language, reusable content, authoring guidance, and structured fields.

FDA's CDER Study Data Standards Research and Development work also reflected this direction. CDER identified the submission of Structured Protocol Information as one of the use cases for testing more modern, standards-based exchange methods for study data and related clinical study content. That work placed structured protocol information within a broader regulatory effort to move beyond document-based and file-based exchange toward more flexible, interoperable representations of clinical study information[1,2].

The ICH M2 discussions around a potential standard protocol template show that by 2017, the industry and regulators were already asking questions that would later become central to ICH M11: What problem could be solved by an ICH action? What content and structure should be harmonized? How should business process, content, structure, and technology relate? What dependencies exist between the protocol and other clinical trial documents?

This history matters because M11 did not emerge in isolation. It followed years of work by standards organizations, regulators, industry groups, and technology communities seeking to improve protocol structure, reuse, and downstream processing. The computable protocol vision was already present. What was missing was global alignment and practical implementation at scale.

The computable protocol is therefore worth revisiting now because M11 changes the implementation context: it does not make protocols computable by itself, but it provides a harmonized foundation that earlier efforts lacked.

[1] The author and former FDA/CDER colleague Dr. Armando Oliva prepared a 2014 presentation entitled *The Computable Protocol*, reflecting early FDA/CDER thinking on how structured protocol information could support review, downstream data capture, protocol-to-analysis alignment, and more systematic examination of study design choices.

[2] The contributions noted here are illustrative rather than exhaustive. The field has benefited from the work of many researchers, standards bodies, and industry groups whose efforts are not individually cited.

3. What M11 Adds

3.1 A Harmonized Framework for Protocol Content

M11 is often described as a clinical protocol template or guideline. That description is accurate, but it understates the broader significance of the work. M11 should also be understood in light of its original concept. The endorsed 2018 M11 Concept Paper stated that "the clinical protocol describes the processes and procedures directing the conduct and analysis of a clinical study." It also recognized the need to support consistency in both structured and unstructured protocol content and to use a technical specification to facilitate electronic exchange.

That framing is important. From the outset, M11 was not limited to creating a more consistent narrative document. It anticipated a relationship between human-readable protocol content, structured protocol content, and electronic exchange. The guideline and template support consistency in how protocol information is developed and presented. The technical specification creates the pathway for that information to be exchanged and reused in more computable forms.

M11 provides a globally harmonized framework for organizing and describing the core components of a clinical trial protocol. It supports consistent representation of study design elements such as objectives, endpoints, estimands, eligibility criteria, interventions, study arms, schedules of activities, analysis populations, and other key protocol content.

This matters because consistency is a prerequisite for reuse, comparison, interoperability, and computational processing. Information that is represented differently in every protocol is difficult to compare. It is difficult to reuse. It is difficult to connect to downstream systems. It is difficult to evaluate systematically.

3.2 M11 as a Framework, not a Complete Computable Solution

M11 does not by itself create a computable protocol. Nor does it automatically enable trial simulation, synthetic population generation, design evaluation, or regulatory review tools. Those capabilities require additional models, data, software, governance, and methods. What M11 provides is a common foundation for organizing protocol information across sponsors, regions, and regulatory settings.

That foundation becomes more useful when connected to technical models that can represent selected study design information as structured data. Models such as the CDISC Unified Study Definitions Model (USDM) provide a way to represent study design elements that can be exchanged, reused, and connected to downstream systems. This is especially important for protocol components such as the Schedule of Activities, where a harmonized protocol structure is useful, but downstream computable use may require more explicit representation of visits, activities, timing, assessments, and relationships among protocol elements. Controlled terminology and semantic metadata can help preserve meaning across systems and use cases.

This does not require treating M11, USDM, controlled terminology, and semantic metadata as a single required architecture. The important point is whether structured protocol content can be represented, exchanged, validated, traced to its source, and reused for study design and execution activities.

In combination, these elements create the possibility of moving from a protocol that is organized in a standard way to a protocol that can support data flow, comparison, visualization, scenario testing, and downstream configuration across the trial lifecycle.

3.3 Implications for Protocol Authoring

The transition is not only technical. It also changes how protocols are authored. Clinical scientists and medical writers have historically worked in narrative documents, where structure is flexible, context can be explained in prose, and revisions can be made through familiar document workflows. Object-based authoring introduces a different discipline. A protocol element may be both narrative content and a reusable data object. An endpoint, eligibility criterion, visit, assessment, intervention, or analysis population may need to be written in a way that is understandable to humans and also sufficiently structured for downstream use.

This is a meaningful workflow change for protocol development teams. Medical writers, clinicians, statisticians, data standards experts, CRO contributors, and technology teams will need to work differently. Organizations will need to decide which protocol elements must be structured, how much narrative flexibility should be preserved, and how structured content will be reviewed, approved, and reused.

3.4 From Standardization to Computability

The value of M11 is that key elements of study design can be represented consistently to support better comparison, traceability, and assessment of design choices. The stronger value proposition lies in deploying the computable version of the protocol, not merely producing a document that follows the M11 structure.

A PDF that follows the M11 structure may improve consistency and readability. However, a computable M11 protocol can do more. It can become a structured source of study design information that supports downstream reuse, comparison, validation, visualization, scenario testing, amendment impact assessment, and traceability.

In this sense, M11 is part of a longer evolution. Narrative protocols could be read. Standardized protocols could be organized more consistently. Structured protocols could be exchanged and reused. Computable protocols can support more systematic examination of study design choices, dependencies, assumptions, and amendment impacts before first patient first visit.

4. Evaluating Trial Design Choices

Protocol standardization is often described in terms of consistency, completeness, and harmonization. These are important benefits. The larger opportunity is the ability to evaluate key aspects of study design before major operational commitments are made.

Consider eligibility criteria. Inclusion and exclusion criteria define the study population, influence recruitment timelines, affect study costs, and shape the applicability of study results. Today, protocol teams review these criteria manually and often conduct separate feasibility assessments to understand recruitment implications. In a more structured environment, eligibility criteria can be evaluated as a coherent set of study requirements. Protocol teams can assess whether criteria are unnecessarily restrictive, identify inconsistent requirements, compare alternative population definitions, and better understand the operational implications of design choices.

The same principle applies to the Schedule of Activities. The schedule defines what procedures are performed, when they occur, and how frequently participants interact with study sites. It is one of the primary drivers of participant burden, site workload, study complexity, and cost. Yet protocol teams frequently evaluate these factors through manual review and experience rather than systematic analysis. More structured protocol information creates opportunities to assess visit burden, identify operational bottlenecks, evaluate the cumulative impact of assessments, and understand the practical implications of design decisions.

The opportunity also applies to endpoint and assessment alignment. Objectives, endpoints, assessments, and visit schedules need to be aligned. Statistical analyses depend on the availability, timing, and quality of collected data. When these relationships are expressed only in narrative text, inconsistencies may be difficult to detect. When they are represented structurally, they become easier to evaluate.

Protocol amendments provide another important example. A seemingly simple modification to an endpoint, eligibility criterion, assessment schedule, or treatment strategy can affect numerous study artifacts and processes, including statistical analysis plans, case report forms, data review procedures, monitoring plans, study reports, site training materials, and regulatory submission components. Today, identifying these impacts is often a labor-intensive exercise involving multiple functional groups.

The value is not the standard itself. The value lies in making design tradeoffs, operational risks, and downstream dependencies visible early enough to be examined before the protocol is finalized and the trial is set in motion.

5. Making Design Assumptions Explicit

Every clinical trial is built upon assumptions. Some are scientific; others are operational. Some relate to recruitment, data collection, statistical analysis, patient retention, site performance, or study execution. Collectively, these assumptions determine whether a trial can answer the questions it was designed to address.

One value of a structured protocol is that these assumptions become easier to see. In a narrative protocol, assumptions may be dispersed across objectives, endpoints, eligibility criteria, assessments, visit schedules, statistical considerations, and operational procedures. Protocol development teams have historically relied on prior experience, expert judgment, feasibility assessments, and cross-functional review to identify potential weaknesses in study design. These approaches remain essential, but they are limited when the underlying protocol information is not structured for comparison, traceability, or systematic analysis. A structured protocol does not eliminate the need for expert judgment, but it makes the underlying logic of the design more visible.

Recruitment assumptions depend heavily on eligibility criteria, site capabilities, disease prevalence, competing trials, and patient willingness to participate. Small changes to inclusion or exclusion criteria can have substantial effects on recruitment feasibility, study duration, and population characteristics. A structured protocol makes it easier to evaluate eligibility criteria as an integrated component of study design rather than as isolated text.

Assessment and endpoint assumptions determine whether the study will collect the right information at the right time and with sufficient quality to support meaningful analysis. Protocol teams routinely make assumptions regarding endpoint timing, assessment schedules, data availability, and participant compliance. More structured protocol information creates opportunities to examine the relationships among objectives, endpoints, assessments, and visit schedules.

Statistical assumptions are also central. Sample size calculations depend on assumptions regarding treatment effects, variability, event rates, and participant retention. Analysis plans depend on assumptions regarding data availability, endpoint timing, protocol adherence, and missing data. More structured representations of study design create opportunities to improve traceability among protocol objectives, endpoints, estimands, analysis populations, and planned analyses.

Operational assumptions are often less visible but equally important. A visit schedule assumes that sites can perform required procedures within available resources and that the enrolled participants are able and willing to adhere to the protocol schedule and required procedures. Data collection requirements assume that investigators can collect information consistently. Monitoring strategies assume that critical data and critical processes have been appropriately identified. Recruitment plans assume that sites have access to suitable participants.

Cost and timeline assumptions are also embedded in protocol design. Each additional visit, procedure, assessment, data collection requirement, or endpoint may affect site workload, participant burden, CRO budgets, monitoring effort, database timelines, and the ability to meet development milestones. A structured protocol creates a better foundation for connecting design choices to their expected operational, time, and cost implications before those choices are locked into the final protocol.

The purpose of evaluating design assumptions is not to eliminate uncertainty. Clinical research will always involve uncertainty. The goal is to make assumptions visible enough to be questioned, tested, documented, and revised while the protocol is still being developed.

6. Testing Design Assumptions Before the Trial Begins

Once design assumptions are visible, they can be examined more systematically. The practical value of a computable protocol becomes more apparent when those assumptions can be explored through what-if analysis, simulation, and scenario testing.

Clinical development teams routinely ask what-if questions during protocol development. What if the eligibility criteria are narrowed? What if the dosing interval changes? What if an endpoint is assessed later? What if the visit window is expanded? What if an interim analysis is added? What if an additional covariate is included in the primary model? What if a procedure is removed to reduce participant burden? These questions are not new. Experienced clinicians, statisticians, pharmacologists, data managers, and clinical operations teams have always considered them during study planning. Trial simulation has also been used for many years to evaluate design assumptions, statistical operating characteristics, dosing strategies, and recruitment scenarios.

The difference is not that M11 makes these methods new. It does not make trial simulation or synthetic population generation new, nor does it make simulation methods more valid by itself.

The difference is that a computable protocol can connect these analyses more directly to the protocol elements they are intended to test. In a document-based environment, protocol text must first be interpreted by humans and translated into separate analytical models, feasibility assessments, simulations, spreadsheets, or operational plans. Each translation step creates friction and introduces opportunities for assumptions to be lost, inconsistently applied, or poorly documented.

A more structured protocol environment changes the starting point. If eligibility criteria, endpoints, visit schedules, dosing rules, analysis populations, estimands, and assessment timing are represented in a consistent and computable manner, then what-if analysis can be tied more directly to the protocol under development. The protocol alone does not answer every question. It provides a structured description of the proposed study design. To evaluate consequences, that structured design must still be connected to other information sources, such as prior study data, historical recruitment experience, disease-specific assumptions, statistical models, synthetic patient populations, and operational performance data.

For example, if a team changes an inclusion criterion, the effect could be evaluated against a synthetic patient population derived from relevant prior studies or real-world data. The team could estimate how many potential participants would remain eligible, how the baseline population might shift, how recruitment timelines might change, and whether the revised population remains aligned with the study objective.

If a dosing interval changes, the structured protocol could identify the affected visits, PK sampling windows, safety monitoring requirements, drug accountability processes, site instructions, and data collection forms. The clinical pharmacologist could evaluate whether the revised schedule remains adequate for exposure-response assessment, while the clinician and safety team could evaluate whether additional monitoring is needed.

If the statistician is considering alternative statistical models, covariates, endpoints, or interim analysis strategies, the computable protocol can provide the structured design context. The statistician could visualize the relationships among objectives, endpoints, estimands, analysis populations, visit schedules, and assessment timing. When connected to historical study data, synthetic patient datasets, and simulation tools, the statistician could compare alternative approaches while the design is still being developed.

In this model, the computable protocol becomes the reference point for what-if analysis. It links protocol design decisions to the data, assumptions, models, simulations, and downstream artifacts needed to evaluate their consequences.

7. Different Questions, One Structured Protocol

The value of M11-based protocol content will not be realized simply because structured data exist behind the document. Users need to be able to see and work with study design information in ways that match their responsibilities. The near-term goal is more practical: structured views that allow different disciplines to examine the same protocol from their own perspectives.

Those views might include the study schema, arms, epochs, visits, procedures, endpoints, eligibility criteria, dosing rules, analysis populations, and assessment windows. They do not need to be elaborate to be useful. A clinician may focus on whether the population, endpoints, safety monitoring, and assessment schedule support the clinical question. A statistician may focus on the alignment among objectives, estimands, endpoints, analysis populations, covariates, missing data assumptions, and visit timing. A clinical pharmacologist may focus on dosing, sampling, exposure-response objectives, food effects, dose adjustments, and the timing of relevant assessments.

Other functions would approach the same structured protocol differently. A data manager may need to determine how protocol requirements translate into data collection forms, edit checks, CDASH and SDTM data structures, and data review procedures. A clinical operations lead or CRO study team may focus on whether the visit schedule is executable by sites and acceptable to participants. Technology and implementation teams may focus on how structured protocol content is represented, exchanged, versioned, and integrated with authoring, EDC, CTMS, data standards, and submission systems. A monitor may need to identify critical data, critical processes, and areas of protocol-driven operational risk. A regulator may focus on whether the study design is adequate to support the conclusions the sponsor intends to draw.

The same structured protocol can support all of these questions, but not by presenting every user with the same static document. The value lies in allowing different users to work from the same underlying study design while seeing the relationships most relevant to their responsibilities.

Over time, more interactive tools may become possible. Users may be able to navigate from a high-level schematic to detailed protocol elements, examine downstream impacts, and evaluate questions such as: What changes if this eligibility criterion is narrowed? What happens if the primary endpoint moves from Week 24 to Week 36? Which visits are affected if the dosing interval changes? What downstream artifacts would need to be updated?

Those capabilities are plausible, but they should be treated as a staged evolution rather than a near-term assumption. They depend on structured content quality, reliable data connections, validated tools, governance, and user trust. If the underlying protocol content is incomplete, inconsistent, or only superficially structured, the interface will not solve the problem.

This is a practical implication of M11 and computable protocol development. The value is not that every user sees the same protocol in the same way. The value is that every user works from the same structured study design while asking the questions that matter for their role.

8. Structured Protocol Libraries and Regulatory Review

8.1 Comparing Protocols Across Prior Studies

The value of structured protocol information increases when protocols can be compared across a library of prior studies. Sponsors may be able to compare a protocol in development against their own prior studies, published trials, consortium resources, or external data sources. This can help sponsor teams ask whether the proposed design resembles prior studies, differs from them, or introduces new assumptions that require justification.

Regulators may have a broader cross-program perspective. Regulatory statisticians and clinicians may have access to prior protocols, submitted study data, review experience, and outcomes across sponsors. Where legally available, appropriately governed, and protected by confidentiality controls, structured protocols and related data could support more consistent and evidence-informed review.

A regulatory reviewer could compare a proposed protocol with prior protocols in a similar indication. A clinical reviewer could examine population definitions, endpoint timing, safety monitoring, visit schedules, and treatment strategies. A statistical reviewer could examine estimands, analysis populations, covariates, multiplicity strategies, interim analyses, and missing data approaches. The purpose would not be to force uniformity. Sponsors may have valid scientific reasons to choose different endpoints, eligibility criteria, statistical approaches, or dosing strategies. The value lies in making differences more visible and easier to evaluate.

8.2 Linking Protocol Comparisons to Study Data

If structured protocols are linked to associated study data, these comparisons become more informative. The reviewer is not merely comparing documents. The reviewer can evaluate proposed design choices in the context of observed performance from prior studies, subject to appropriate governance, confidentiality protections, and scientific judgment.

This could support better sponsor-regulator dialogue during protocol development. Instead of relying only on narrative review and individual memory, reviewers could identify relevant precedents, highlight potential design risks, and ask more targeted questions about the sponsor's assumptions. Regulators may also be able to identify recurring design issues, amendment patterns, endpoint challenges, feasibility problems, or statistical assumptions across multiple sponsors and development programs.

Synthetic or simulated patient populations may further extend this capability. Prior study data could be used, under appropriate governance and privacy protections, to create synthetic populations that preserve relevant statistical characteristics without exposing individual patient records. These populations could then be used to test eligibility criteria, endpoint timing, missing data assumptions, recruitment expectations, dosing scenarios, and statistical approaches.

This would not replace the need for the actual trial. Nor would it transform prior data into new confirmatory evidence. Rather, it would strengthen the ability to test design choices during protocol development and review.

A related future direction is the use of structured protocol content to support more timely signals during trial conduct. If protocol-defined endpoints, assessments, visit schedules, safety monitoring requirements, and decision thresholds are represented structurally, they could eventually help support predefined alerts, operational monitoring, and sponsor-regulator dialogue during study execution. That use would require clear governance, validation, access controls, and agreement on how such signals should be interpreted.

8.3 Governance and Appropriate Use

This capability must be governed carefully. Sponsors may be concerned if cross-sponsor data are used to challenge a proposed design without transparency regarding the basis for comparison. Regulators will need to consider how such analyses are framed, how confidentiality is protected, how precedents are interpreted, and how to avoid creating the appearance that prior designs define a required standard.

Structured protocol libraries should not become a mechanism for forcing uniformity. Their value lies in supporting better questions, not automatic conclusions. There is also a risk that structured protocols create a new surface for regulatory scrutiny. Sponsors may worry that more structured content will lead to more questions, more comparisons, and more requests for justification. That concern should not be dismissed. A more transparent protocol can make assumptions easier to evaluate, and that may reveal issues that were previously harder to see.

The goal should be better review, not heavier review. Structured protocol content should support more focused dialogue about the most important design assumptions, not create an expectation that every structured element will become a point of regulatory challenge. A cross-sponsor structured protocol library would also require legal authority, secure data infrastructure, access controls, data governance, and staff with the clinical, statistical, regulatory, and computational expertise needed to use such a resource responsibly. In resource-constrained regulatory environments, these requirements are substantial.

The concept is therefore best understood as a direction of travel rather than an immediately available capability. Structured protocol content can improve regulatory review over time, but realizing the full cross-sponsor learning opportunity will require institutional investment, governance, and clear policies for appropriate use.

Several governance issues would need to be addressed:

- Access controls. Prior sponsor data should be used only by authorized regulatory personnel for appropriate review purposes.
- Query transparency. If a regulator raises a concern based on cross-sponsor experience, the sponsor should understand the nature of the comparison, even if confidential details of other sponsors' studies cannot be disclosed.
- Scientific appropriateness. A protocol should not be compared with prior studies simply because they share a broad indication. Meaningful comparison requires attention to disease stage, population, mechanism of action, endpoint definition, assessment timing, control arm, standard of care, geography, and other design features.

- Evidentiary status. Comparisons with prior studies, synthetic patient testing, and what-if analyses can inform protocol review and discussion, but they do not replace evidence generated by the clinical trial itself.

- Sponsor explanation. Sponsors should have the opportunity to explain why their design differs from prior approaches. The purpose of structured regulatory comparison should not be to enforce historical conformity, but to make design assumptions, deviations from precedent, and potential risks easier to discuss.

Governance must also protect innovation. If structured protocol libraries are used only to identify how a proposed design differs from prior studies, they could unintentionally create pressure toward conformity. Clinical development often advances because a sponsor proposes a design that differs from precedent: a different endpoint, a more targeted population, a novel estimand, a new dosing strategy, or an adaptive design feature.

A cross-sponsor regulatory reference library should therefore be used to support better questions, not to create a regression to the mean. A design that differs from prior studies should not be treated as deficient simply because it is different. The relevant question is whether the difference is scientifically justified, whether the assumptions are clear, and whether the design can support the conclusions the sponsor intends to draw.

In this sense, governance must protect both confidentiality and novelty. Confidentiality protects sponsors from inappropriate disclosure or misuse of prior submitted data. Protection of novelty helps ensure that structured comparison does not become a barrier to innovative trial design. The goal should be governed learning from prior experience, not automatic precedent-setting.

9. From Structured Protocol to Trial Operations

For sponsors and CROs, the practical issue is operational use. Once protocol content is represented as structured data, the key question is how that approved representation is used by the systems and teams responsible for planning, conducting, monitoring, analyzing, and submitting the study.

A structured protocol representation can also serve as an authoritative source for computable protocol content. This does not eliminate the need for a human-readable protocol document, but it can reduce the repeated reinterpretation of protocol information across authoring, operations, data management, statistics, monitoring, and submission workflows. The operational value comes when downstream teams use the approved structured representation rather than rebuilding protocol content manually from PDFs, Word documents, spreadsheets, and local trackers.

This requires more than an M11-formatted document. Sponsors will need authoring tools, data models, validation rules, change control, staff training, and integration with existing trial systems. The business case becomes stronger when structured protocol content supports practical use cases such as protocol-to-SAP alignment, Schedule of Activities review, downstream system configuration, amendment impact assessment, CRF generation, data standards mapping, or structured sponsor-CRO handoff.

CROs are central to this transition. In many trials, CRO teams operationalize the protocol across sites, systems, monitoring activities, data collection, and study conduct. If the sponsor creates a structured protocol but the CRO must recreate or reinterpret that information manually, much of the value is lost. The goal should be to exchange the authoritative structured protocol representation in a form that downstream teams and systems can consume, version, and update when the protocol changes.

This makes operating model decisions important. Sponsors and CROs need to define which version of the structured protocol is authoritative, how it is exchanged, which downstream systems use it, how amendments are reflected, and who is responsible for validating that the structured content matches the approved protocol. The PDF or document view remains necessary for communication and submission. The computable version is what enables reuse, comparison, automation, impact analysis, and more reliable downstream execution.

10. Structured Protocol Governance

For computable protocols to become reliable operational assets, sponsors and CROs will need governance models that go beyond traditional document control. A structured protocol creates two related but distinct representations: the human-readable protocol and the structured computable content behind it. Both must remain aligned. If the narrative text says one thing and the structured field says another, organizations need a clear rule for resolving the discrepancy. In most cases, the approved protocol document will remain the legal and regulatory artifact, but the structured representation may increasingly drive downstream processes. That makes reconciliation essential.

Sponsors will also need to define ownership of structured protocol content. Medical writing may own the narrative protocol, but structured elements may also be used by clinical operations, CRO partners, data management, biostatistics, clinical pharmacology, safety, regulatory operations, and technology teams. Without clear ownership, structured protocol content risks becoming everyone’s dependency and no one’s accountability.

Governance should address practical questions about ownership, validation, exchange, change control, and downstream use:

- Who is responsible for creating and validating the structured protocol content?
- Which organization or function owns the authoritative structured protocol representation?
- How is structured content exchanged between sponsor and CRO systems?
- Who approves changes to structured fields?
- How are changes to endpoints, eligibility criteria, visits, assessments, dosing, estimands, or analysis populations propagated to downstream systems?
- How are structured protocol versions reconciled with protocol amendments?
- What happens if a downstream system uses structured protocol content that later changes?
- How are assumptions, what-if analyses, and simulation outputs linked to the protocol version that informed them?
- How are sponsor and CRO responsibilities documented when structured protocol content is used for operational execution?

These questions are not administrative details. They determine whether computable protocol content can be trusted for study execution, analysis planning, amendment impact assessment, and regulatory review. A mature computable protocol environment will require version control, role-based approvals, audit trails, change impact tracking, and clear reconciliation between narrative and structured representations. It will also require sponsor-CRO governance that defines responsibility for structured content across handoffs, amendments, systems, and operational execution. Without that governance, structured protocol content may create new operational and regulatory risks even as it enables new efficiencies.

11. Protocol Amendments as a Near-Term Use Case

Protocol amendments provide a practical near-term business case for computable protocols. Amendments are common, costly, and disruptive. They require review, approval, communication, implementation, site retraining, system updates, document updates, and often changes to data collection, monitoring, statistical planning, and regulatory submissions. They can also create periods in which sites operate under different protocol versions, increasing complexity and execution risk.

Published benchmark data show that the prevalence of protocols with at least one amendment has increased substantially, and that the mean number of amendments per protocol has also increased. A recent Tufts CSDD benchmark study examined 950 protocols and 2,188 amendments. It reported that the prevalence of protocols with at least one amendment increased from 57% to 76%, while the mean number of amendments per protocol increased from 2.1 to 3.3. The same research found that the average time from identifying the need to amend to last oversight approval was 260 days. Earlier Tufts work (2016) reported median direct costs of $141,000 for a substantial Phase II amendment and $535,000 for a substantial Phase III amendment.

This matters for M11 because computable protocol content provides a practical mechanism for identifying some avoidable problems earlier. The amendment business case is strongest when it focuses on avoidable amendments, not amendment elimination. A computable protocol should not be promoted as a tool that will prevent all amendments. Some amendments reflect responsible learning during development, emerging safety information, changes in external knowledge, competitive context, or regulatory expectations.

The more credible claim is that structured protocol content may help reduce amendments caused by preventable design weaknesses and may shorten the impact assessment for amendments that remain necessary. A computable M11-based protocol could support earlier detection of these issues by enabling eligibility criteria testing, Schedule of Activities burden assessment, endpoint-to-assessment alignment checks, protocol-to-SAP consistency checks, downstream impact analysis, identification of missing or inconsistent data collection requirements, and assessment of whether planned visits, procedures, and assessments are operationally feasible.

The business case does not require claiming that computable protocols will eliminate amendments. The practical case is that structured protocol content may help reduce avoidable amendments, shorten amendment impact assessment, and improve the ability to implement necessary amendments consistently. For sponsors and CROs, the near-term case rests on those operational gains, not on claims of full automation or amendment elimination.

12. Staged Adoption and Near-Term Tools

The long-term vision of an interactive protocol design environment is important, but the near-term opportunity is more practical and incremental. Early tools are likely to be focused and task-specific: protocol consistency checkers, Schedule of Activities burden visualizers, eligibility criteria evaluators, protocol-to-SAP alignment tools, amendment impact tracers, endpoint-to-assessment mapping tools, structured protocol comparison tools, and tools that support sponsor-CRO handoff of structured protocol content.

Adoption will also depend on whether structured authoring feels like a practical extension of protocol development rather than rigid data entry. If structured authoring preserves only the appearance of structure, computability will remain limited. If it makes protocol development harder without producing useful downstream benefits, study teams will have little reason to change established workflows.

These tools require well-structured protocol elements, clear rules, traceability, and expert review. They can be implemented incrementally and evaluated against practical measures such as reduced rework, earlier issue detection, improved cross-functional alignment, faster amendment impact assessment, better documentation of design rationale, and more reliable downstream implementation.

The path forward should therefore be staged:

1. Structure the protocol.
2. Validate high-value use cases.
3. Connect structured protocol elements to downstream systems and data.
4. Establish governance for traceability, validation, sponsor-CRO handoff, amendment management, and regulatory acceptability.
5. Introduce advanced analytics, simulation, and interactive tools where bounded, validated use cases show clear value.

This staged approach is important because few organizations will transform protocol development all at once. More likely, sponsors and CROs will begin with focused use cases where the value is clear: reducing amendment burden, improving protocol-to-SAP alignment, assessing participant burden, supporting CRF generation, tracing protocol changes to downstream impacts, or improving structured handoff from sponsor design to CRO execution.

The long-term vision matters, but near-term credibility depends on solving practical problems.

13. Accountable Use of Computable Protocol Tools

As computable protocol tools become more capable, accountability becomes more important. Structured protocols, simulation tools, synthetic data, protocol comparison tools, and interactive review environments can support protocol development, but only if they are used with appropriate controls.

The core value of the computable protocol comes from the structured representation of study design. Without structured protocol content, downstream tools must rely on manual interpretation of narrative text. That may be useful in limited settings, but it does not solve the larger problems of traceability, consistency, reuse, or systematic evaluation of design decisions. With structured protocol content, tools can help users navigate relationships among protocol elements, compare design alternatives, summarize downstream impacts, identify inconsistencies, and connect protocol elements to relevant data, assumptions, or prior studies.

For example, a statistician might ask: "If the Week 24 endpoint moves to Week 36, what protocol elements and analysis assumptions are affected?" A structured review environment could identify the endpoint definition, Schedule of Activities, visit windows, analysis population, missing data assumptions, sample size assumptions, and SAP sections likely to be affected. If connected to simulation tools, it could help initiate scenario analyses. If connected to a structured protocol library, it could identify comparable prior studies. If connected to submitted study data in a regulatory environment, it could support comparisons with observed outcomes from similar designs.

These tools would not determine the correct answer or replace the statistician, clinician, pharmacologist, regulator, CRO expert, or study team. Human experts remain accountable for evaluating scientific rationale, clinical relevance, statistical validity, operational feasibility, and regulatory acceptability.

Validation also needs to be matched to the intended use of the tool. Organizations will need to distinguish among content validation, system validation, and analytical or model validation. A structured content checker, an authoring platform, and a simulation tool raise different validation questions. An output that influences a study design decision cannot be treated as an informal suggestion if it becomes part of the design rationale. The organization must be able to show what data were used, what protocol version was analyzed, what assumptions were applied, what output was generated, who reviewed it, and how the decision was made.

This requires provenance, audit trails, version control, documentation of use, and clear decision rights. The more directly a tool influences protocol design, statistical assumptions, data collection requirements, operational decisions, or regulatory submissions, the more important that validation infrastructure becomes. Appropriate controls are especially important when structured protocols are connected to prior study data, synthetic populations, simulation outputs, or cross-sponsor regulatory reference libraries. Assumptions must be explicit, data sources must be appropriate, synthetic data must be distinguished from real clinical evidence, and results must be traceable to protocol elements. Protocol versions, simulation inputs, model outputs, and design decisions must be managed so that the rationale for changes can be reconstructed.

These controls are not barriers to innovation. They are what make innovation credible in a regulated environment. In such an environment, a study team could begin with an M11-based protocol structure and progressively build a study design. As eligibility criteria are added, the team could evaluate recruitment implications. As endpoints are defined, the team could assess alignment with assessments, visit schedules, estimands, and analysis populations. As dosing rules are added, the team could evaluate safety monitoring, PK sampling, drug accountability, and site burden. As visit schedules are developed, the team could evaluate participant burden and operational feasibility.

The protocol would remain a regulatory and operational document, but it would also become a structured design asset. The immediate goal is improved visibility, consistency, traceability, and accountable use of structured protocol content.

14. Conclusion

The computable protocol vision has existed for more than twenty years. The industry has long understood that protocols could be more than static documents. ICH M11 now provides the globally harmonized framework that earlier efforts lacked.

M11’s value lies in making protocol content more consistent, structured, and connected. When that content is linked to an underlying data model, study teams can examine design assumptions, test alternatives, identify dependencies, and understand amendment impacts before the trial is set in motion.

The goal is not simply a better-formatted protocol. The goal is a protocol that can support clearer study design, more reliable downstream execution, and better traceability from design choices to trial conduct. The promise of the computable protocol is straightforward: know more before the trial begins.

References

CDISC. Protocol Representation Model (PRM) v1.0. Published 30 January 2010. https://www.cdisc.org/standards/foundational/prm/prm-v1-0.

CDISC. Digital Data Flow and Unified Study Definitions Model resources. https://www.cdisc.org/ddf.

Getz K, Smith Z, Botto E, Murphy E, et al. New benchmarks on protocol amendment practices, trends and their impact on clinical trial performance. *Therapeutic Innovation & Regulatory Science*. 2024;58:539-548. doi:10.1007/s43441-024-00622-9.

Getz K, Stergiopoulos S, Short M, et al. The impact of protocol amendments on clinical trial performance and cost. *Therapeutic Innovation & Regulatory Science*. 2016;50(4):436-441. doi:10.1177/2168479016632271.

International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. Final Concept Paper: ICH M11 Clinical electronic Structured Harmonised Protocol (CeSHarP). Dated 14 November 2018; endorsed by the Management Committee 15 November 2018. https://database.ich.org/sites/default/files/M11_EWG_Concept_Paper.pdf.

International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. ICH Harmonised Guideline: Clinical Electronic Structured Harmonised Protocol (CeSHarP) M11. Final Version. Adopted 19 November 2025. https://database.ich.org/sites/default/files/ICH_Step4_M11_Final_Guideline_2025_1119.pdf.

International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. M11 Technical Specification: Clinical Electronic Structured Harmonised Protocol. Final Version. Adopted 19 November 2025. https://database.ich.org/sites/default/files/ICH_Step4_M11_Final_TechnicalSpecification_2025_1119.pdf.

Kahn MG. Computable protocol models for interactive trial design. *Drug Information Journal*. 2002;36:487-497. doi:10.1177/009286150203600302.

National Institutes of Health and U.S. Food and Drug Administration. NIH-FDA Phase 2 and 3 IND/IDE Clinical Trial Protocol Template. https://grants.nih.gov/policy-and-compliance/policy-topics/clinical-trials/protocol-template.

TransCelerate BioPharma Inc. Common Protocol Template and Clinical Content & Reuse assets. https://www.transceleratebiopharmainc.com/initiatives/clinical-content-reuse/.

U.S. Food and Drug Administration. CDER Study Data Standards Research and Development. Content current as of 9 December 2014. https://www.fda.gov/drugs/electronic-regulatory-submission-and-review/cder-study-data-standards-research-and-development.